

vision transformers

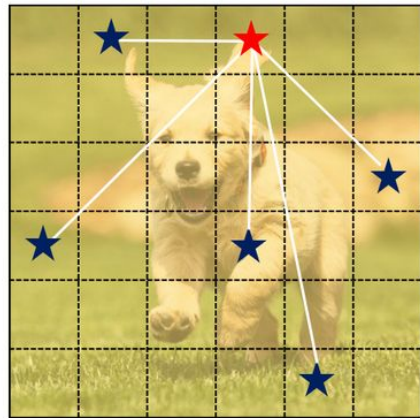
& training one to read pediatric x-rays



what is a vision transformer?

a neural network that reads a picture the way a language model reads a sentence:

- ❖ it chops the image into small patches
- ❖ treats each patch like a word
- ❖ uses attention to work out how the patches relate

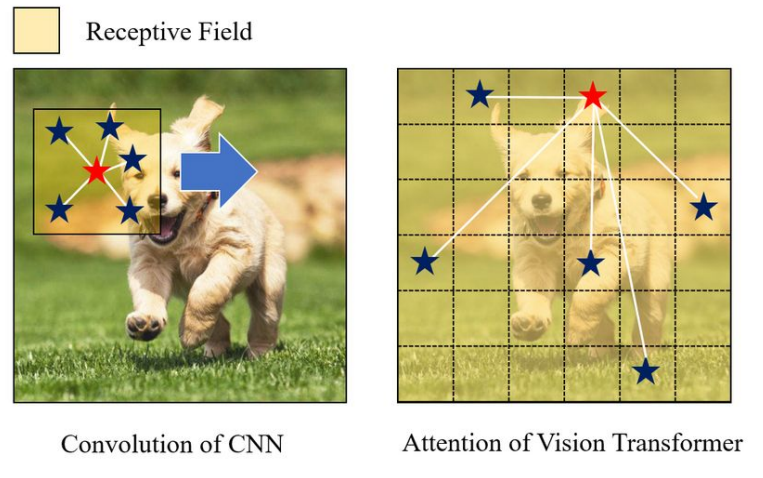


Attention of Vision Transformer



how do they work?

- ❖ step 1: split the image into patches
- ❖ step 2: create query, key, and value vectors + compute attention scores
- ❖ step 3: weighted combination



A CNN builds up from small local pieces, while a ViT relates the whole image at once!

ViTs struggle heavily on small or specialized datasets. They require massive pre-training, and high computational power.

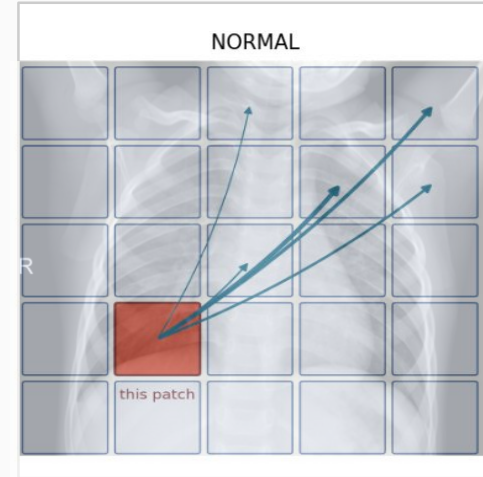


step 1: split the image into patches

instead of looking at pixels directly, a ViT divides an image into patches (for example, 16×16 pixels each)

an x-ray might become:
patch 1 = upper left lung
patch 2 = upper right lung
patch 3 = heart region

each patch is then converted into a vector
called an embedding



Chest X-Ray Images (Pneumonia)
dataset from Kermany et al. (2018)



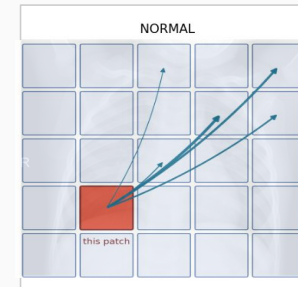
step 2: create query, key, and value vectors + compute attention scores

for every patch embedding, the model creates three new vectors, and computes attention scores

query = what am i looking for
key = what information do i contain
value = what information should i send

$$\text{softmax}\left(\frac{\text{Q} \times \text{K}^T}{\sqrt{d_k}}\right) \text{V}$$

= Z



Attention determines which image patches should influence one another.

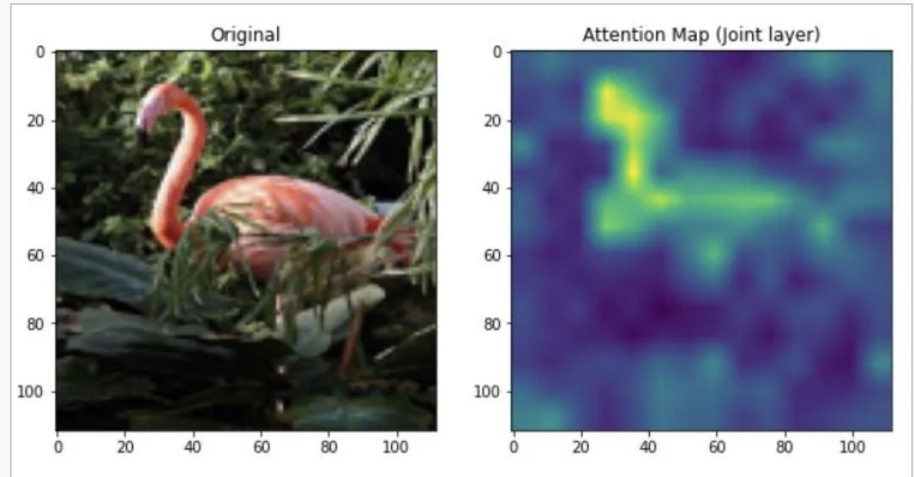
The thicker the arrows, the higher the computed attention value.



step 3: weighted combination

after calculating which image patches are most important, the model creates a weighted combination

it combines information from all the patches, but gives more influence to the patches it thinks matter most





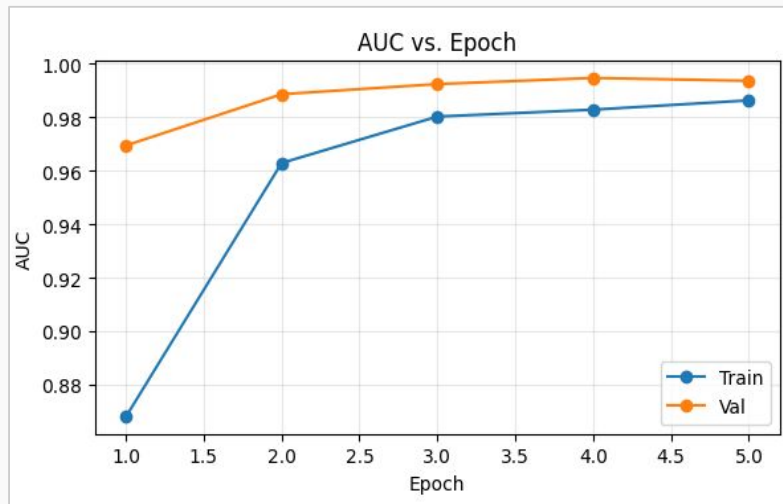
my results

i began with a ViT already trained on 14 million everyday images, then fine-tuned it on my x-ray dataset (transfer learning)

86% accuracy

0.93 AUC

0.92 Pneumonia Recall

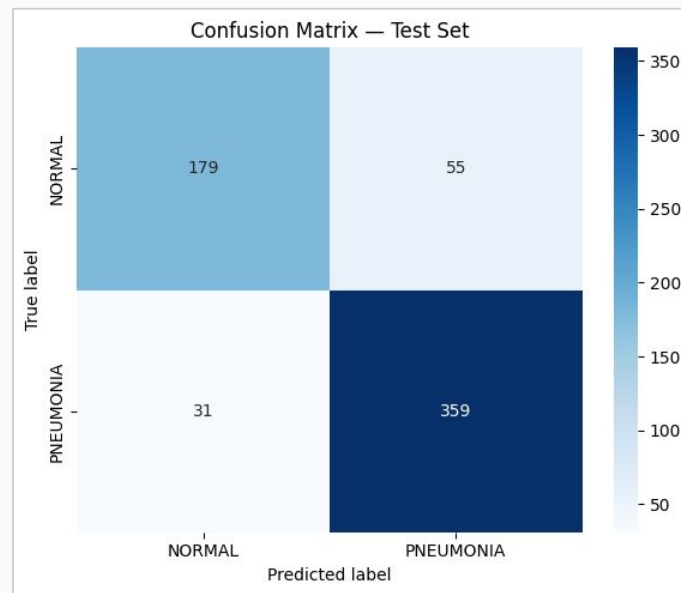




my results

caught 359 of 390 pneumonia cases,
missing only 31

flagged 55 of 234 normal x-rays as
pneumonia (false positives)





references

Baek, S., Park, J., Vepakomma, P., & Kim, S.-L. (2022). *Visual transformer meets CutMix for improved accuracy, communication efficiency, and data privacy in split learning* (Preprint). ResearchGate.

<https://www.researchgate.net/publication/361733806>